# Strategic Imagination Capacity Under Chronic Material Pressure: A Dynamic State-Space Model

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### Abstract

Chronic material pressure can constrain not only the objective opportunity structure available to individuals but also their capacity to generate, compare, elaborate, and revise alternative courses of action. Existing approaches often risk conflating objective opportunity, perceived opportunity, behavioral execution, and the cognitive capacity to represent alternatives. This paper develops a dynamic state-space model that separates these dimensions and specifies how they may evolve over time.

The model is organized around two analytically distinct state variables: objectively audited opportunity structure, $O^{objective}$, and Strategic Imagination Capacity (SIC). SIC is defined as the capacity to generate, elaborate, compare, and revise representations of future alternative courses of action under active constraint. Perceived opportunity, rehearsal, recovery/cognitive availability, and observable execution are treated as distinct measurement or process variables rather than as substitutes for either state variable.

The original sequential three-stage conception is reformulated as a dynamic state space. Stage 1 represents a narrow but real opportunity margin with preserved SIC; Stage 2 represents partial erosion of comparison capacity; Stage 3A represents substantial SIC erosion while objective opportunity may remain open; and Stage

3B represents objective opportunity closure with comparatively preserved SIC. A joint closure configuration, $J$, represents simultaneous low objective opportunity and low SIC without constituting a separate stage.

The paper specifies hypotheses concerning closure exposure, rehearsal, recovery, individual sensitivity, objective/perceived opportunity dissociation, behavioral equifinality, recovery trajectories, and temporal persistence. A longitudinal, multi-source observational design is proposed with four principal measurement waves over 18 months and higher-frequency measurement of closure exposure and process variables. A Bayesian hierarchical dynamic model is specified together with simulation-based power analysis, kernel comparison, measurement validation, and pre-specified decision rules. As of the present version, the simulation stage of the protocol has been executed in full — five simulation experiments covering identifiability, parameter recovery, recovery boundaries, statistical power, and state-classification validity — and its results are reported in Section 7.13.

The model does not claim that chronic material pressure inevitably produces strategic closure, nor does the design establish causal effects, population prevalence, or persistence beyond the observation window. Its primary contribution is a testable structural separation between objective opportunity, represented opportunity, strategic capacity, process exposure, recovery conditions, and observable execution.

**Keywords:** strategic imagination; material pressure; opportunity structure; state-space models; chronic scarcity; longitudinal design; cognitive availability; strategic planning; behavioral execution.

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## 1. Introduction

Chronic material pressure is commonly described in terms of insufficient income, restricted access to resources, unemployment, unstable employment, housing insecurity, constrained mobility, or limited institutional access. Such conditions can directly reduce the set of actions that an individual can realistically undertake. They may also alter the temporal and cognitive conditions under which future alternatives are generated and evaluated. A central difficulty arises when these two effects are treated as equivalent.

An individual may face a genuinely restricted opportunity structure while retaining a substantial capacity to imagine and compare alternative futures. Conversely, an individual may face objectively available opportunities while having difficulty generating, elaborating, comparing, or revising possible courses of action. The observable behavior may be similar in both cases: little or no execution. This creates a basic identification problem. Observing that

$$Execution_t \approx 0$$

does not by itself establish whether the limiting condition is primarily $O_t \downarrow$ or $SIC_t \downarrow$. The distinction matters because the two configurations imply different theoretical trajectories and different empirical measurements.

This paper therefore asks: *how does chronic material and objective opportunity closure relate to the dynamics of Strategic Imagination Capacity, and under what conditions do rehearsal and recovery alter that relationship?*

The question is deliberately narrower than a general theory of poverty, motivation, cognition, or behavior. The objective is to specify a dynamic representation in which objective opportunity and the capacity to represent and compare alternatives are analytically distinct and empirically testable. The paper introduces

Strategic Imagination Capacity (SIC) as a behavioral construct referring to the ability to (1) generate multiple alternatives, (2) elaborate those alternatives into concrete courses of action, (3) compare alternatives across relevant dimensions, and (4) revise alternatives when constraints change. This definition differs from optimism, motivation, aspiration, intelligence, or self-reported hope: it concerns an observable capacity to manipulate representations of possible future action.

The paper proceeds through four conceptual shifts. First, a previously sequential model of erosion is reformulated as a dynamic state-space model, in which the stages become reference regions rather than mandatory temporal steps. Second, objective opportunity, perceived opportunity, SIC, rehearsal, recovery, and execution are operationally separated. Third, the model specifies explicit longitudinal hypotheses rather than relying on descriptive stage narratives. Fourth, decision rules are defined before empirical analysis, including a smallest effect of practical interest $\epsilon$, a minimum important difference for opportunity dissociation, and a diagnostic equivalence margin for behavioral classification. The resulting framework is intended to function simultaneously as a theoretical model, an empirical protocol, and a basis for reproducible simulation and analysis.

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## 2. Conceptual Background

### 2.1 Chronic Material Pressure and Constrained Choice

Material scarcity can influence behavior through multiple mechanisms. Limited resources can reduce the feasible set of actions, increase the cost of mistakes, restrict mobility, and compress available time. Research on scarcity and cognitive bandwidth has further suggested that resource constraints can influence attention and decision-making. However, a reduction in available opportunity is not equivalent to a reduction in the capacity to represent alternatives.

The distinction is particularly important under chronic rather than acute constraint. Chronic pressure can create repeated exposure to blocked pathways, failed attempts, administrative barriers, unstable income, and restricted mobility. The theoretical possibility examined here is that repeated closure exposure may eventually become associated with changes in the capacity to generate and compare alternatives. The model does not assume that this process is inevitable or universal; it treats it as an empirical question.

### 2.2 Objective and Perceived Opportunity

Opportunity has at least two analytically distinct forms. The first is the set of courses of action that are independently verifiable as materially and institutionally accessible, $O_t^{objective}$. The second is the set of courses of action that an individual currently represents as available and executable, $O_t^{perceived}$. These sets need not coincide: a person may fail to represent an objectively available opportunity, and conversely an apparently available opportunity may not be objectively actionable because of eligibility requirements, geography, direct cost, institutional barriers, timing, or other constraints. The distinction therefore cannot be resolved by self-report alone.

### 2.3 Strategic Imagination Capacity

SIC is defined here as *the capacity to generate, elaborate, compare, and revise representations of future alternative courses of action under active material constraint.* The construct contains four analytically separable dimensions:

$$SIC = (Generate, Elaborate, Compare, Revise)$$

Generation concerns the production of distinct alternatives; elaboration concerns the specification of concrete requirements and steps; comparison concerns the ability to evaluate alternatives across multiple dimensions; and revision concerns the ability to modify an existing plan after the introduction of a new constraint. The construct is behavioral rather than purely self-reported.

### 2.4 Rehearsal and Recovery

The model distinguishes capacity from process exposure. Rehearsal refers to repeated engagement with generating, reviewing, comparing, and revising alternatives, $R_t$, whereas SIC refers to the underlying capacity to perform these operations, $SIC_t$. Similarly, recovery refers to conditions that restore cognitive availability; it is not defined as a latent psychological trait. Two components are distinguished: $G^{objective}$, representing available unstructured time, and $G^{subjective}$, representing internal quiescence or perceived cognitive availability. A behavioral component may provide a third source of measurement.

### 2.5 Execution

Execution is an observable behavioral outcome:

$$Execution_t = f(O_t, SIC_t, X_t)$$

where $X_t$ represents additional execution constraints not captured by the two principal state variables. Thus $Execution \neq SIC$ and $Execution \neq O$. Non-execution can occur under both $(O \downarrow, SIC \uparrow)$ and $(O \uparrow, SIC \downarrow)$ — this is the basis for the behavioral equifinality hypothesis developed later (H3).

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## 3. Dynamic State-Space Model

### 3.1 From Sequential Stages to a Dynamic State Space

The original three-stage conception could be interpreted as a necessary sequence, $1 \rightarrow 2 \rightarrow 3$. That interpretation is too restrictive. The revised model uses two state variables: $O_t$, the objective/material/institutional opportunity structure, and $SIC_t$, Strategic Imagination Capacity. The state of an individual at time $t$ is therefore represented as $\mathcal{S}_t = (O_t, SIC_t)$, and the two dimensions may change at different rates and need not move in the same direction.

Execution is not treated as a third state dimension; it remains $Execution_t = f(O_t, SIC_t, X_t)$, with rehearsal, recovery, and closure exposure treated as process variables affecting transitions between states. A constitutive distinction is also maintained between $O_t^{objective}$ and $O_t^{perceived}$: the latter is not simply a noisy substitute for the former.

### 3.2 Four Reference Regions

**Stage 1 — Narrow but Real Margin.** Chronic material pressure exists, but $SIC_t$ is relatively preserved and $O_t^{objective} > 0$. The actionable opportunity space is constrained but non-zero.

**Stage 2 — Erosion of Comparison Mechanism.** SIC is partially impaired, particularly in the ability to compare or revise alternatives. Objective opportunity may remain restricted or partially open; Stage 2 therefore does not imply a specific objective opportunity level.

**Stage 3A — Internal SIC Erosion.** $SIC_t \downarrow$ while $O_t^{objective}$ may remain low, moderate, or relatively open. 3A therefore describes a low-SIC region rather than a unique fixed point.

**Stage 3B — Objective Opportunity Closure with Preserved SIC.** $O_t^{objective} \downarrow$ while $SIC_t$ remains comparatively preserved. This configuration demonstrates why inability to act should not automatically be interpreted as inability to imagine alternatives.

**Joint Closure Configuration.** $J = (O \downarrow, SIC \downarrow)$ is a region of joint closure rather than a separate stage.

### 3.3 Dynamic SIC Equation

The conceptual dynamic is:

$$\frac{dSIC_i}{dt} = -\alpha_i E_t + \beta R_t + \gamma G_t$$

where $E_t$ is cumulative closure exposure, $R_t$ is rehearsal, $G_t$ is recovery/cognitive availability, and $\alpha_i$ is the estimated individual sensitivity to closure exposure. The parameter $\alpha_i$ is not assumed to be a pre-existing resilience trait; it is estimated from longitudinal trajectories.

### 3.4 Closure Exposure

Closure exposure is represented as a temporally weighted accumulation:

$$E_t = \int_0^t K(t-\tau)\, S_{closure}(\tau)\, d\tau$$

Candidate kernels include the exponential, $K_E(\Delta t) = e^{-\lambda \Delta t}$, and the power law, $K_P(\Delta t) = (1+\Delta t)^{-\rho}$. The kernel comparison is limited to the observed temporal window: an 18-month study cannot establish persistence over decades.

### 3.5 Possible Trajectories

The model permits several trajectories: gradual erosion, $1 \to 2 \to 3A$; direct external closure, $1 \to 3B$; closure followed by SIC erosion, $3B \to 3A$, treated probabilistically as $P(3B \to 3A) = f(E_t, R_t, G_t, \theta_i)$ rather than assumed inevitable; recovery, $3A \to 2 \to 1$; reopening with preserved SIC, $3B \to 1$; and joint closure, $3B \to J$ or $3A \to J$.

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## 4. Hypotheses

The hypotheses translate the state-space model into empirically testable claims. They are probabilistic rather than deterministic.

**H1 — Closure-to-Erosion.** Among individuals entering the study with preserved SIC and objectively restricted opportunity, greater cumulative closure exposure is associated with subsequent SIC decline: $\alpha_0 > 0$. H1 applies specifically to the baseline configuration corresponding to 3B.

**H1a — Rehearsal Moderation.** Rehearsal attenuates the negative association between closure exposure and SIC: $\partial SIC/\partial E < 0$ and $\partial^2 SIC/\partial E\,\partial R > 0$.

**H1b — Recovery Threshold.** Recovery may operate conditionally rather than as a purely additive process. A threshold specification requires $G_t > \tau_G$ for a meaningful protective effect; the empirical comparison is between additive and threshold/interaction models.

**H1c — Closure × Rehearsal Intensification.** The protective effect of rehearsal may increase as cumulative closure exposure rises: $\partial^2 SIC/\partial E\,\partial R > 0$. The primary test evaluates $\delta_1 + 2\delta_2 E^*_{ref} > 0$.

**H1d — Individual Sensitivity.** Individuals differ in their estimated sensitivity to closure exposure, $\alpha_i = \alpha_0 + b_i$; the variance $\sigma^2_\alpha$ is therefore an estimand.

**H2 — Objective/Perceived Opportunity Dissociation.** Objective and perceived opportunity are not necessarily equivalent, $O_t^{objective} \neq O_t^{perceived}$. The empirical test concerns non-equivalence under the specified measurement framework.

**H3 — Behavioral Equifinality of Non-Execution.** Comparable non-execution may occur under $(O \downarrow, SIC \uparrow)$ and $(O \uparrow, SIC \downarrow)$. Therefore behavioral non-execution alone should not be sufficient to distinguish 3A from 3B.

**H4 — Differential Recovery Pathways.** Following opportunity reopening, 3B cases may transition toward Stage 1 without an intermediate SIC decline, while 3A cases may show progressive SIC recovery. The distinction is evaluated longitudinally rather than inferred from a single behavioral observation.

**H5 — Kernel Persistence.** Alternative temporal-memory specifications produce different fits to the observed relationship between closure exposure and SIC. The study compares exponential and power-law kernels, with flexible alternatives only if simulation demonstrates sufficient identification. The hypothesis is bounded by the observed temporal window.

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## 5. Operationalization

The study distinguishes $O^{objective}$, $O^{perceived}$, $SIC$, $R$, $G$, and $Execution$ as separate constructs, each defined before the instrument used to measure it.

**5.1 Objective Opportunity.** Objective opportunity is *the set of courses of action independently verifiable as materially and institutionally accessible to a given individual within a defined time window and constraint set*, $O_{it}^{objective} = \{o_1, \ldots, o_n\}$, across the domains of employment, education, housing, mobility, financing, and care. Actionability requires consideration of factual availability, eligibility, application feasibility, requirements, geography, direct cost, institutional access, and temporal window. Objective opportunity is independently audited rather than treated as metaphysically exhaustive.

**5.2 Perceived Opportunity.** Perceived opportunity is the set of courses of action represented by the individual as currently available and executable, $\Delta O = O^{objective} - O^{perceived}$. For vector-valued opportunity, a pre-specified distance is used, $D(O^{objective}, O^{perceived}) = \sum_d w_d \, |O_d^{objective} - O_d^{perceived}|$, with weights fixed in advance. Perceived opportunity is measured before the SIC task to reduce contamination by task exposure.

**5.3 Strategic Imagination Capacity.** SIC is measured behaviorally through a structured scenario task in which participants generate alternatives, elaborate them, identify requirements and constraints, compare alternatives, and revise an alternative after a new constraint. The four dimensions are initially scored separately; a composite SIC score is created only if factor analysis supports a common structure. Parallel forms A–D reduce practice effects, and potential contamination from general verbal fluency and working memory is addressed through control measures.

**5.4 Rehearsal.** Rehearsal is *repeated engagement with generating, reviewing, comparing, or revising alternative courses of action*: $R_t$ is process/exposure, whereas $SIC_t$ is capacity. Measurement uses behavioral logs and brief repeated probes.

**5.5 Recovery / Cognitive Availability.** Recovery is represented by objective unstructured time, subjective internal quiescence, and a behavioral restoration/attention measure. Alternative specifications are compared: $G = aT + bC$ versus $G = aT + bC + c(T \times C)$ versus a conditional threshold specification. The threshold applies to the effect of recovery, not to the definition of recovery itself.

**5.6 Execution.** Execution is observable behavioral conversion of an available or represented alternative into action (application, registration, contact, training enrollment, or a defined pathway step), measured independently from opportunity and SIC.

**5.7 Individual Sensitivity.** Individual sensitivity is estimated as $\alpha_i = \alpha_0 + b_i$; the correct inferential direction is to estimate $\alpha_i$ in order to explain variation, not to measure a putative resilience trait and assume it equals $\alpha_i$.

**5.8 Kernel / Temporal Memory.** The study measures closure monthly and SIC at four principal waves, $t_0, t_6, t_{12}, t_{18}$, with closure exposure computed as $E_{it} = \int_0^t K(t-\tau)\, S_{closure}^{audit}(\tau)\, d\tau$. No inference is made about persistence beyond the observation window.

**5.9 Joint Closure.** $J = (O\downarrow, SIC\downarrow)$ is not a separate instrument or stage.

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## 6. Study Design

Three claims constrain the design: $O$ and SIC are distinct state variables; their positions can change over time; and their measurement must remain analytically separate. The design is therefore longitudinal, multi-source, and temporally hybrid, with within-individual SIC change as the primary inferential target.

**6.1 Study Type.** The study is a longitudinal repeated-measures observational study with principal waves at $t_0, t_6, t_{12}, t_{18}$ and monthly closure measurement used to construct temporal exposure. The observational design is deliberate: direct manipulation of opportunity closure in a population experiencing chronic material pressure would raise ethical and practical problems, so the design prioritizes longitudinal observation and measurement separation rather than experimental manipulation.

**6.2 Target Population.** Adults experiencing chronic material pressure, operationalized through sustained low or unstable income over the preceding 12 months, restricted objective opportunity in at least three prespecified domains, and absence of an acute crisis that would dominate the measurement period. Participants must have preserved SIC at baseline for H1; participants with lower baseline SIC may be retained for recovery analyses such as H4.

**6.3 Eligibility for H1.** Eligibility requires $SIC_{i,0} > \tau_{SIC}$ and $O_{i,0}^{objective} \leq \tau_O$, with thresholds established using an independent calibration procedure or pre-registered external normative procedure; the main

sample is not used to define its own thresholds. The smallest effect of practical interest, $\epsilon > 0$, is fixed before data collection using independent calibration, a criterion-referenced procedure, or a measurement-error framework, and represents the smallest standardized $\alpha_0$ considered substantively consequential.

**6.4 Sampling.** Stratified purposive sampling may recruit through community organizations, vocational/adult education, and referral/snowball pathways, with potential stratification on gender, urban/rural location, income source, and household composition.

**6.5 Observation Schedule.** At principal waves: SIC, objective opportunity, perceived opportunity, rehearsal, recovery, and execution. Monthly: closure detection, exposure verification, and time-diary components. Continuous or weekly: rehearsal logs and execution records where feasible.

**6.6 SIC Measurement.** The SIC task uses parallel forms, each containing alternative generation, elaboration, comparison, and revision after constraint introduction. Coding is performed independently where feasible, and inter-rater reliability is assessed.

**6.7 Closure Exposure.** Monthly participant reporting serves as event detection, not as the final closure measure. Potentially consequential events are independently verified using administrative records, program rosters, third-party confirmation, or structured assessor verification. Only verified events enter $S^{audit}_{closure}(\tau)$, so that $E_t = \int_0^t K(t-\tau)\, S^{audit}_{closure}(\tau)\, d\tau$.

**6.8 Opportunity Audit.** Objective opportunity is audited at principal waves, with a random subsample independently assessed by a second assessor to estimate inter-rater reliability.

**6.9 Rehearsal and Recovery.** Rehearsal is monitored using brief repeated logs. Recovery includes monthly time diaries, subjective cognitive availability, and a behavioral restoration task at principal waves; at least one non-self-report component is required.

**6.10 State Classification.** Two classification systems are maintained. Analytical reference regions — Stage 1, Stage 2, 3A, 3B, and J — are descriptive regions in $(O, SIC)$ and are not necessarily mutually exclusive. Hypothesis-testing categories, used where tests require 3A versus 3B, are defined as $3A : SIC \leq \tau_{SIC}$ and $3B : SIC > \tau_{SIC} \wedge O^{objective} \leq \tau_O$, with priority rules fixed in advance.

**6.11 Statistical Model.** The primary dynamic model is:

$$SIC_{i,t+1} = \phi\, SIC_{it} - \alpha_i E_{it} + \beta R_{i,t:t+1} + \gamma G_{i,t:t+1} + \delta_1 (E_{it} R_{i,t:t+1}) + \delta_2 (E^2_{it} R_{i,t:t+1}) + \eta (E_{it} G_{i,t:t+1}) + u_i + \varepsilon_{it}$$

with individual exposure sensitivity $\alpha_i = \alpha_0 + b_i$, $b_i \sim N(0, \sigma^2_\alpha)$; random intercept $u_i \sim N(0, \sigma^2_u)$; and residual $\varepsilon_{it} \sim N(0, \sigma^2_\varepsilon)$. Here $\phi$ is the autoregressive coefficient, $\alpha_0$ is the population mean closure sensitivity, and $\sigma^2_\alpha$ is the between-person heterogeneity. Priors are weakly informative and fixed before analysis.

**6.12 Simulation-Based Power.** Power is evaluated through Monte Carlo simulation across candidate sample sizes $N = 200, 300, 400, 500, 600$, varying ICC, variance ratios, attrition, the $E \to SIC$ effect, $E \times R$, $\sigma^2_\alpha$, kernel misspecification, and $\delta_2$. The result is a power surface rather than a single claimed minimum sample size. Section 7.13 reports the executed results of this simulation.

**6.13 Missingness and Attrition.** Attrition may be informative. Primary analysis assumes MAR with sensitivity analysis under MNAR; retention procedures include flexible follow-up modalities, repeated contact, and participant-centered scheduling. Multilevel/time-aware imputation or direct Bayesian missing-outcome modeling may be used where appropriate.

**6.14 Identification Limits.** The design supports within-person association between closure exposure and SIC, moderation by rehearsal and recovery, longitudinal recovery patterns following opportunity reopening, and comparison of temporal-memory specifications. It does not by itself establish strong causal E→SIC effects, population prevalence, or persistence beyond 18 months.

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## 7. Analysis Plan

Analysis follows measurement → state construction → exposure construction → dynamic model → hypothesis testing. No post hoc hypothesis creation is permitted within confirmatory analysis.

**7.1 Data Assembly.** Four principal SIC waves produce three transitions per participant. Closure exposure is calculated only from events occurring before or at the relevant transition time, $E_{it}^{\leq t} = \int_0^t K(t-\tau)\, S_{closure}^{audit}(\tau)\, d\tau$; future closure events are never incorporated into earlier exposure values.

**7.2 Measurement Validation.** Pre-specified adequacy criteria are:

| Criterion | Adequate | Marginal | Inadequate |
|---|---|---|---|
| SIC $\omega$ | ≥ .70 | .60–.69 | < .60 |
| SIC inter-rater ICC | ≥ .75 | .60–.74 | < .60 |
| Opportunity audit $\kappa$ | ≥ .70 | .50–.69 | < .50 |
| Detection sensitivity | ≥ .70 | .60–.69 | < .60 |

Inadequate measurement prevents the affected variable from supporting confirmatory hypothesis testing.

**7.3 State Variables.** Objective opportunity is retained as a domain-level vector. SIC is treated as a composite only if factor analysis supports a common construct. Perceived/objective divergence is analyzed using the pre-specified vector distance of Section 5.2.

**7.4 Exposure Construction.** Two principal kernels are constructed, $E_{it}^{exp} = \int e^{-\lambda(t-\tau)} S_{closure}^{audit}(\tau)\, d\tau$ and $E_{it}^{power} = \int (1+t-\tau)^{-\rho} S_{closure}^{audit}(\tau)\, d\tau$, with exposure standardized before interaction modeling.

**7.5 Primary Dynamic Model.** Bayesian hierarchical estimation using HMC is primary; frequentist mixed-effects estimation using REML is a sensitivity analysis. If convergence fails, model reduction follows a pre-specified sequence: (1) full model, (2) remove $\delta_2$, (3) remove random exposure slope, (4) simplify kernel, (5) simpler fixed-kernel model. Convergence problems are documented rather than concealed through arbitrary prior changes.

**7.6 Hypothesis Tests.**

*H1.* Let $CI(\alpha_0)$ be the 95% credible/confidence interval and $\epsilon > 0$ the pre-specified smallest effect of practical interest. Because H1 is directional: **supported** if $\inf CI(\alpha_0) > \epsilon$; **no evidence of a practically important association** if $CI(\alpha_0) \subseteq [-\epsilon, \epsilon]$; **inconclusive** if the interval intersects $[-\epsilon, \epsilon]$ but is not contained within it. The three categories are mutually exclusive and exhaustive, preventing statistical distinguishability from zero from being confused with substantive importance.

*H1a.* Test $\delta_1 > 0$; a positive interaction indicates attenuation of the negative closure-SIC relationship by rehearsal.

*H1b.* Compare additive and threshold recovery specifications, with threshold interaction $G \cdot \mathbf{1}(T > \tau_T, C > \tau_C)$, testing $H_0 : \gamma_2 = 0$ against $H_1 : \gamma_2 > 0$; model comparison uses predictive criteria such as LOO-CV or WAIC.

*H1c.* The marginal effect of rehearsal is $\partial SIC/\partial R = \beta + \delta_1 E + \delta_2 E^2$, with cross-partial $\partial^2 SIC/\partial E\,\partial R = \delta_1 + 2\delta_2 E$. The primary reference point is $E^*_{ref}$, and the primary test is $\delta_1 + 2\delta_2 E^*_{ref} > 0$; the full E-range is reported descriptively.

*H1d.* Heterogeneity is tested through $\sigma^2_\alpha$. Because variance is constrained to be non-negative, posterior mass relative to a pre-specified practical region is preferred over a conventional zero-crossing test. Individual estimates are treated as hierarchical shrinkage estimates rather than stable individual causal slopes. As documented in Section 7.13, this test carries an explicit design-based testability boundary in $\sigma_\alpha$.

*H2.* Using $D(O^{objective}, O^{perceived}) = \sum_d w_d |O^{objective}_d - O^{perceived}_d|$, a minimum important difference is fixed before analysis; H2 concerns whether observed divergence exceeds the pre-specified practical-equivalence region.

*H3.* Two classifiers are compared — behavior-only and the full state-variable model — with the principal comparison $\Delta_{AUC} = AUC_{full} - AUC_{behavior\text{-}only}$, alongside cross-validated AUC, balanced accuracy, calibration, Brier score, and confusion matrix. A pre-specified $\Delta^*_{AUC}$ defines the practical diagnostic difference.

*H4.* Only reopening events meeting the pre-specified minimum event count are analyzed, comparing event-time SIC trajectories. Claims of no intermediate decline require an equivalence/non-inferiority margin rather than simple failure to reject a null hypothesis; claims of progressive recovery require multiple post-event measurements.

*H5.* Exponential and power-law kernels are compared using predictive performance. A kernel is supported only if the predictive comparison favors it by the pre-specified criterion and parameter estimates remain within meaningful non-boundary regions. Interpretation remains restricted to the observed 18-month window; Section 7.13 reports the executed identifiability check.

**7.7 Missing Data.** MAR is primary. Multiple imputation or direct Bayesian missing-outcome modeling is used according to the multilevel structure of the data, with MNAR pattern-mixture sensitivity analyses assessing robustness.

**7.8 Sensitivity Analysis.** Sensitivity analyses include threshold variation, SIC composite versus dimensions, alternative recovery specifications, alternative kernels, exclusion of low-confidence exposure intervals, MAR/MNAR assumptions, and outlier sensitivity.

**7.9 Multiple Testing.** Primary hypotheses are H1, H1a, H1c, H1d, H2, H4, H5; secondary hypotheses are H1b and H3. Exploratory analyses include domain-level and wave-specific analyses. Benjamini–Hochberg FDR control is applied separately within primary and secondary families.

**7.10 Simulation.** Monte Carlo simulation evaluates parameter recovery, power, kernel identification, interaction detection, variance-component recovery, and attrition effects, including scenarios in which $\delta_2$ is independently varied. If a hypothesis is demonstrably underpowered by the pre-specified design, it is not rescued through post hoc model changes.

**7.11 Falsification Logic.** Each hypothesis has a pre-specified disconfirming region:

| Hypothesis | Disconfirming pattern |
|---|---|
| H1 | effect interval lies within practical-null region |
| H1a | no positive moderation |
| H1b | threshold model does not improve prediction |
| H1c | cross-partial does not become positive at $E^*_{ref}$ |
| H1d | negligible heterogeneity |
| H2 | opportunity divergence lies within practical-equivalence region |
| H3 | behavioral-only model performs comparably within the pre-specified margin |
| H4 | predicted trajectory distinction is absent or untestable |
| H5 | kernels cannot be distinguished within the observation window |

**7.12 Reporting Limits.** The analysis does not infer causality, prevalence, clinical diagnosis, intervention efficacy, or persistence beyond the observation period.

**7.13 Simulation Results (v0.2.0-simulation).** The simulation pipeline specified in Sections 6.12 and 7.10 has been executed in full, producing five experiments with a traceable provenance chain from data-generating process through CSV artifacts to a verified results archive. The headline results are:

| Experiment | Result |
|---|---|
| H5 identifiability | 16 cells evaluated; minimum kernel correlation 0.3691, maximum 0.9974; status: testable in identified regions of the $(\lambda, \rho)$ grid only |
| Parameter recovery | 8 of 8 primary model parameters recovered with bias below 1% at $N = 300$ |
| Recovery boundary (H1d) | $\sigma_\alpha$ recovery adequate at $\sigma_\alpha \geq 0.10$, transitional at 0.05, marginal at 0.03, inadequate at 0.01 |
| Power surface | Minimum $N$ for 80% power: H1 = 100, H1a = 100, H1c = 200, H1d = 100 |
| State scenarios | 4 of 4 validation checks passed (3B, 3A, J, and the priority rule of Section 6.10) |

These results carry two consequences for the confirmatory analysis plan. First, H5 (Section 7.6) is not uniformly identifiable across the full $(\lambda, \rho)$ parameter space: the exponential and power-law kernels can be distinguished in some regions of the grid and not others, so H5 is reported as testable only within the distinguishable regions rather than treated as testable or untestable as a whole. Second, H1d (Section 7.6) carries a design-based testability condition tied to the magnitude of $\sigma_\alpha$: if the posterior interval for $\sigma^2_\alpha$ is compatible with $\sigma_\alpha < 0.05$, H1d is reported as **not testable within the observed window**, and not as "not supported" — a distinction the recovery-boundary simulation shows is necessary because recovery of small variance components is itself inadequate at this design's sample sizes and wave count, independent of whether the true heterogeneity is zero. A related finding is that power for H1c decreases as $\sigma_\alpha$ increases while power for H1d increases, an interaction between parameter heterogeneity and the detectability of the intensification term that is treated as a substantive design finding rather than a defect to be tuned away.

All simulation-derived figures above have been cross-checked against their underlying CSV artifacts through an automated verification pass; no empirical participant data is included in this stage.

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## 8. Discussion

The discussion interprets findings only within the inferential limits established by the design.

**8.1 What the Design Can Establish.** If supported by adequate measurement, the study can identify within-person closure/SIC associations, moderation by rehearsal and recovery, objective/perceived opportunity divergence, behavioral diagnostic insufficiency, differential longitudinal recovery patterns, and temporal-memory differences within the observation window. It cannot independently establish causal mechanisms.

**8.2 Interpretation of H1.** If $CI(\alpha_0)$ lies entirely above $\epsilon$, the result supports an association of practically important magnitude between cumulative closure exposure and subsequent SIC decline among the defined baseline population. If $CI(\alpha_0) \subseteq [-\epsilon, \epsilon]$, there is no evidence of a practically important association under the study's resolution. If the interval is wide and intersects both the practical-null region and practically important values, the result is inconclusive — an inconclusive result is a design finding, not evidence that the phenomenon does not exist.

**8.3 H1a–H1c.** H1c is interpreted conditionally on H1a. If both are supported, rehearsal moderation is consistent with increasing protection at higher closure exposure. If H1a is supported but H1c is not, moderation may exist without evidence of intensification. If H1a is unsupported, H1c is reported but is not interpreted as evidence for a meaningful protective-barrier pattern.

**8.4 H2.** H2 concerns empirical non-equivalence under the specified measurement framework. Observed divergence could reflect differences in representation, measurement error, SIC-related processes, or objective opportunity concealment or complexity; the design does not independently adjudicate among these interpretations.

**8.5 H3.** If behavior-only classification performs substantially worse than the full state-variable model, the result supports the claim that non-execution alone does not contain sufficient information to distinguish 3A from 3B. This is a statement about information and measurement, not a demonstration of different causal mechanisms.

**8.6 H4.** If opportunity reopens and 3B cases move toward execution without an intermediate SIC decline while 3A cases exhibit repeated SIC recovery, this would be consistent with SIC functioning as a partially independent state variable rather than simply reflecting contemporaneous opportunity. This does not establish statistical or causal independence.

**8.7 H5.** If a power-law specification fits better than an exponential specification, the observed relationship is more consistent with slower decay among the compared forms over the study's observation window. This does not establish that a power-law mechanism is universally correct, and — per Section 7.13 — the comparison itself is only meaningful within the regions of parameter space where the simulation confirms the two kernels are distinguishable.

**8.8 What the Results Do Not Say.** The model does not imply that chronic material pressure inevitably causes SIC erosion; that SIC is the only determinant of execution; that 3A is a clinical condition; that $\alpha_i$ is equivalent to resilience or grit; that the model describes all populations; that reopening itself is a proven causal mechanism; or that observed patterns persist beyond the study window.

**8.9 Unsupported or Inconclusive Results.** Null and inconclusive results remain informative. A failure to support H1 may indicate that the association is smaller than the pre-specified practical threshold, that the study lacks sufficient precision, that the relationship varies across contexts, or that the specified model does not adequately represent the process. Unsupported H1a or H1c would weaken the rehearsal-protection hypothesis without invalidating the state-space architecture. Unsupported H3 would challenge the claim that behavioral information is insufficient. Untestable H4 or H5 represents a limitation of the available event structure or temporal resolution.

**8.10 Model Modification versus Replacement.** Two types of revision must be distinguished. Parameter-level modification allows specific parameter estimates, thresholds, or functional forms to be reconsidered in future studies. Structural modification would be required if the empirical results contradict the architecture itself. Changes motivated by observed results are post hoc model development and are not treated as confirmatory evidence from the original protocol.

**8.11 Future Work.** Future work may include longer observation periods, larger samples, replication across contexts, additional event-time measurements, stronger causal designs, further SIC validation, additional recovery specifications, and alternative temporal-memory models, each tied to unresolved empirical findings rather than introduced independently of them.

**8.12 Closing Interpretation.** The central empirical question is not whether material pressure produces a universal psychological outcome. It is whether objective opportunity structure and strategic capacity exhibit distinguishable trajectories under chronic pressure, and whether those trajectories are systematically related to closure exposure, rehearsal, and recovery.

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## 9. Limitations

**9.1 Design Limitation.** The study is observational and therefore does not establish a causal effect of closure exposure on SIC. This is a deliberate constraint rather than a design error, motivated by the ethical and practical infeasibility of manipulating opportunity closure in the target population. Stronger causal identification would require exogenous or quasi-exogenous variation, a natural experiment, or another design capable of separating exposure from unobserved confounding.

**9.2 Measurement Limitation.** Objective opportunity is independently verifiable under pre-specified criteria but is not necessarily exhaustive. Perceived opportunity is self-reported and may be partially contaminated by SIC; the design mitigates this by measuring perceived opportunity before the SIC task and preserving temporal ordering in analysis. SIC itself remains subject to construct-validity concerns despite behavioral measurement, parallel forms, and control tasks. Residual measurement error may attenuate associations or generate apparent divergence between variables.

**9.3 Identification Limitation.** Potential reciprocal relationships remain, $R_t \leftrightarrow SIC_t$ and $O_t \leftrightarrow SIC_t$; temporal separation improves ordering but does not establish unidirectional causality. With only three SIC transitions per participant, individual estimates $\hat{\alpha}_i$ are heavily shrunk toward the population mean through hierarchical partial pooling and should not be interpreted as stable individual-level causal parameters.

**9.4 Temporal Limitation.** The principal observation window is 18 months, which limits identification of long-tail persistence and prevents extrapolation to years or decades. H4 additionally depends on sufficient opportunity-reopening events and repeated post-event measurements.

**9.5 Generalizability.** The target population consists specifically of adults entering the study with preserved SIC and objectively restricted opportunity, as specified in Section 6.3. The results therefore cannot be in-

terpreted as population prevalence estimates or automatically generalized to all forms of poverty, scarcity, unemployment, or institutional exclusion. Transportability requires replication.

**9.6 State-Classification Limitation.** The underlying model is continuous in $(O^{objective}, SIC)$. Stage 1, Stage 2, 3A, 3B, and J are reference regions, not ontologically discrete states. The mutually exclusive 3A/3B classification used in H3, H4, and the transition matrix depends on pre-specified thresholds and a fixed priority rule; threshold sensitivity analyses are reported, but classification should not be interpreted as evidence that the underlying state space consists of discrete categorical states.

**9.7 Kernel Identification.** Monthly exposure measurement improves temporal resolution but does not by itself guarantee that alternative kernels can be distinguished. As reported in Section 7.13, simulation now confirms that identification is regionally conditional rather than uniform across the $(\lambda, \rho)$ grid; H5 is treated as untestable outside the identified regions, and longer or denser observation would be required for stronger kernel identification across the full space.

**9.8 Attrition.** Attrition may be informative, particularly if participants experiencing greater SIC erosion are more difficult to retain. Retention procedures, MAR modeling, and MNAR sensitivity analyses reduce but do not eliminate this concern; strongly informative attrition could bias estimated trajectories and heterogeneity.

**9.9 Scope of the Model.** Execution is represented as $Execution_t = f(O_t, SIC_t, X_t)$, where $X_t$ represents additional execution constraints not operationalized in the present study. Its role is analytical: it marks the boundary of what the two-state-variable formulation can claim to explain. Future studies measuring specific execution constraints — legal, financial, institutional, or relational — could estimate components of $X_t$ directly.

**9.10 What Would Be Required to Resolve the Remaining Questions.**

| Question | Required evidence |
|---|---|
| Causal E→SIC effect | Exogenous or quasi-exogenous variation |
| Long-term persistence | Observation beyond 18 months |
| Kernel identification | Longer and/or denser exposure history |
| Recovery dynamics | Repeated event-time measurement |
| Generalizability | Replication across populations and contexts |
| SIC construct validity | Independent behavioral and longitudinal validation |
| Reciprocal $O \leftrightarrow SIC$ dynamics | Finer temporal measurement |
| Stable individual sensitivity | More than three transitions per participant |

**9.11 Closing Boundary.** The study is a test of a specified dynamic representation of objective opportunity and Strategic Imagination Capacity under chronic material pressure. It is not a causal theory of poverty or a general theory of human behavior. Its strongest claims are bounded by the observational design, measurement validity, eligibility criteria, the 18-month observation window, and the ability of the data to identify the proposed functional forms.

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## 10. Conclusion

This study develops and specifies a dynamic state-space model for examining Strategic Imagination Capacity (SIC) under chronic material pressure. The model distinguishes two analytically separate state variables — objectively audited opportunity structure, $O^{objective}$, and SIC — and treats perceived opportunity, rehearsal,

recovery, and execution as distinct measurement or process components rather than collapsing them into a single behavioral construct.

The resulting framework replaces a strictly sequential interpretation of strategic erosion with a dynamic representation in which objective opportunity and SIC can move differently over time, permitting the empirical distinction between configurations such as objective closure with preserved SIC and SIC erosion despite remaining objective opportunity. Non-execution is consequently treated as an observable outcome that does not, by itself, identify which state configuration generated it.

The empirical status of the model is determined by the pre-specified analysis plan of Section 7. H1 tests the association between cumulative closure exposure and subsequent SIC change, using a ROPE-based decision rule that distinguishes statistical from practical significance. H1a–H1c examine rehearsal and recovery as moderators; H1d examines heterogeneity in estimated sensitivity, subject to the design-based testability condition confirmed in Section 7.13; H2 tests the empirical non-equivalence of objective and perceived opportunity; H3 tests whether behavioral non-execution alone is diagnostically sufficient to distinguish 3A from 3B; H4 examines differential recovery trajectories following opportunity reopening; and H5 compares alternative temporal-memory specifications within the regions of the parameter space where the executed simulation confirms they are identifiable. Each hypothesis is reported as supported, not supported, inconclusive, or not testable according to the decision rules specified before analysis.

The protocol has now completed its simulation stage: parameter recovery, power, recovery-boundary, kernel-identifiability, and state-classification behavior have all been verified in silico, with the full pipeline, 180 automated tests, and a documented provenance chain from data-generating process to results archive. This closes the simulation stage of the protocol without altering its confirmatory status: no empirical participant data has been collected, and the study does not establish causal effects, population prevalence, long-term persistence beyond the 18-month observation window, or universal applicability of the model.

Its contribution is instead methodological and structural: a formally specified, now simulation-verified way to distinguish objective opportunity, represented opportunity, strategic capacity, process exposure, recovery conditions, and observable execution within a common longitudinal framework. Future research can extend this structure through longer observation periods, repeated event-time measurements, stronger causal identification, replication across populations and institutional contexts, and further validation of the SIC construct — extensions that should be driven by specific unresolved findings rather than by post hoc modification of unsupported predictions.

The central contribution of the study is therefore not a claim that chronic material pressure inevitably produces strategic closure. It is the specification of a testable, now simulation-verified model in which objective opportunity and the capacity to represent and compare alternatives are distinct dimensions whose trajectories may diverge, interact, and recombine over time.

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## References — Initial Working List

*Note: this is a provisional bibliography for the initial manuscript. Exact editions, DOIs, page ranges, and citation formatting should be verified before publication rather than inferred.*

Appadurai, A. Work on the capacity to aspire and the social conditions shaping future-oriented imagination.

Bandura, A. Work on self-efficacy, agency, and human action.

Carver, C. S., Scheier, M. F., and related work on self-regulation and goal processes.

Chetty, R., Hendren, N., Kline, P., and related work on opportunity structures and mobility.

Haushofer, J., & Fehr, E. Work on poverty, scarcity, and economic decision-making.

Kahneman, D. Work on attention, cognitive capacity, and decision-making.

Mani, A., Mullainathan, S., Shafir, E., & Zhao, J. Work on poverty and cognitive function under scarcity.

Mullainathan, S., & Shafir, E. *Scarcity: Why Having Too Little Means So Much.*

Snyder, C. R. Work on Hope Theory, goals, pathways, and agency.

Additional methodological literature on longitudinal mixed-effects models, Bayesian hierarchical models, state-space models, missing-data analysis, and kernel estimation should be incorporated during the final reference audit.

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## Current Manuscript Status

Conceptual architecture: closed · Hypotheses: closed · Operationalization: closed · Study design: closed · Analysis plan: closed · Decision rules: closed after $\epsilon$ correction · Discussion: closed · Limitations: closed · Conclusion: closed · **Simulation: executed (v0.2.0-simulation, 2026-09-26) — 5 experiments completed, ~180 automated tests passing, documentation cross-verified against CSV artifacts** · Pilot: not yet conducted · Empirical data: none yet.

Pending next steps: OSF pre-registration upload; Zenodo DOI registration for the code archive; IRB approval for the pilot study; pilot study ($N \approx 30$–$50$); threshold calibration from an independent sample; main data collection.